Halley's Comet Leaves a Trail of Shooting Stars

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You do not need to wait for Halley's Comet to visit to see its fireworks. This week, Earth passed through a trail of dusty **debris** left behind by the famous comet, and the crumbs put on a show as the Orionid meteor shower reached its peak.

On Oct. 21, **astrophotographer** Tim McCollum captured a spectacular **composite image** above Toccoa, Georgia. Between 3 and 4 a.m., he photographed three vivid Orionid meteors streaking across the sky. Each shooting star appears to race away from the shower's **radiant** the point in the sky the meteors seem to come from which sits near Betelgeuse, the red giant star in the **constellation** Orion. Halley's Comet itself is more than 3 billion miles from Earth right now, but the particles it shed still carve bright paths through the autumn sky when they hit our atmosphere.

McCollum's picture holds more wonders. Orion's Belt glistens beside the glow of the Orion **Nebula**, one of the closest major star-forming regions to Earth. The Rosette **Nebula** also appears, nestled in the dense starfields and dust lanes of the Milky Way, which slants diagonally across the image. Catching a shooting star on camera takes skill, luck and patience. McCollum used a star tracker to keep his camera fixed on the **radiant** and took many 30-second exposures, later combining them into one image with astrophoto software.

If you missed the peak, there is still time: the Orionid shower stays active until Nov. 7. Head outside on a clear night and you might spot one of Halley's dazzling leftovers or even a 'Halloween Fireball' from the Taurid meteor shower, made from **debris** of Comet 2P/Encke.

Word Treasure

debris -- Small pieces of material left behind after something breaks apart or passes by; here, dusty crumbs shed by a comet.

radiant -- The point in the sky that meteors in a shower seem to stream away from.

constellation -- A pattern of stars in the sky that people have given a name and a shape.

nebula -- A huge cloud of gas and dust in space where new stars can form.

composite image -- A picture made by combining many separate photos into one.

astrophotographer -- A person who takes photographs of the night sky and objects in space.

Background you need

A meteor shower happens when Earth passes through a trail of dusty debris left behind by a comet, and the tiny particles burn bright as they hit the atmosphere.

Halley's Comet sheds such crumbs as it travels, so its leftovers can light up the sky even when the comet itself is billions of miles away.

The Orionid shower is named for Orion, because its meteors seem to race away from a point called the radiant near that constellation.

Photographers catch the brief streaks by keeping a camera fixed on the radiant and combining many long exposures into a single composite image.

Break it down

LEAD: Earth passed through dusty debris left by Halley's Comet, putting on the Orionid meteor shower at its peak

KEY EVENT: Astrophotographer Tim McCollum captured three vivid Orionid meteors above Toccoa, Georgia, on Oct. 21

+- WHO: Tim McCollum, astrophotographer

+- WHERE: Above Toccoa, Georgia, with the radiant near Betelgeuse in the constellation Orion

+- HOW: A star tracker held the camera on the radiant while many 30-second exposures were combined into one composite image

EVIDENCE: The composite also shows Orion's Belt, the Orion Nebula, the Rosette Nebula and the Milky Way's dense starfields

OPEN QUESTION: Whether clear skies will let viewers catch more meteors -- or a Taurid 'Halloween Fireball' -- before the shower ends after Nov. 7

Why it matters

Meteor showers let anyone watch space debris burn up in our atmosphere with no equipment at all, connecting everyday skywatchers to a famous comet that is now billions of miles away. They also give photographers a predictable chance to capture the crumbs comets leave behind.

Test what you remember

1. What causes the Orionid meteor shower?

(A) Fireworks launched from Georgia
(B) Stars falling out of the constellation Orion
(C) Dust from the Moon drifting into the air
(D) Earth passing through dusty debris left behind by Halley's Comet

2. When did Tim McCollum photograph the meteors?

(A) Between 3 and 4 a.m. on Oct. 21
(B) During last year's shower
(C) On the night of Nov. 7
(D) At noon on Oct. 21

3. How many vivid Orionid meteors did he photograph streaking across the sky?

(A) Thirty
(B) Seven
(C) Three
(D) One

4. Where do the meteors seem to come from?

(A) A radiant near Betelgeuse in the constellation Orion
(B) The Milky Way's dust lanes
(C) The Orion Nebula
(D) The surface of Halley's Comet

5. How did McCollum create his composite image?

(A) He borrowed a picture from a telescope company
(B) He drew the meteors by hand
(C) He took one single photo at noon
(D) He used a star tracker and combined many 30-second exposures

6. Until when does the Orionid shower stay active?

(A) Until Halley's Comet returns
(B) It ended at its peak
(C) Until Nov. 7
(D) Until Oct. 21

Answer key (try the quiz first!): 1: D 2: A 3: C 4: A 5: D 6: C

Questions to think about

MY THOUGHTS (at least 20 words):